

LAB-1:

Aim: To perform and understand fundamental DBMS operations, including database creation, table management, data manipulation, modification, and retrieval using MySQL.

SHOW databases;

CREATE DATABASE stud1_test;

SHOW DATABASES;

```
MySQL localhost:3306 ssl SQL > CREATE DATABASE stud1_test;
Query OK, 1 row affected (0.0054 sec)
MySQL localhost:3306 ssl SQL > SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| stud1_test |
| sys |
+-----+
```

CREATE DATABASE stud1test;

SHOW DATABASES;

```
MySQL localhost:3306 ssl SQL > CREATE DATABASE stud1test;
Query OK, 1 row affected (0.0064 sec)
MySQL localhost:3306 ssl SQL > SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| stud1_test |
| stud1test |
| sys |
+-----+
```

DROP DATABASE stud1test;

SHOW databases;

```
MySQL localhost:3306 ssl SQL > DROP DATABASE stud1test;
Query OK, 0 rows affected (0.0045 sec)
MySQL localhost:3306 ssl SQL > SHOW databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| stud1_test |
| sys |
+-----+
```

USE stud1_test;

CREATE TABLE Persons (

PersonID int,

LastName varchar(25),

FirstName varchar(25),

Address varchar(100),

City varchar(25)

)

SHOW TABLES;

```
MySQL localhost:3306 ssl stud1_test SQL > SHOW TABLES;
+-----+
| Tables_in_stud1_test |
+-----+
| persons |
+-----+
```

DESCRIBE Persons;

```
MySQL localhost:3306 ssl stud1_test SQL > DESCRIBE Persons;
```

Field	Type	Null	Key	Default	Extra
PersonalID	int	YES		NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
City	varchar(25)	YES		NULL	

DROP TABLE Persons;

CREATE TABLE Customers (CustomerID int,LastName varchar(25),FirstName
varchar(25),Address varchar(100),City varchar(25), primary key
(CustomerID));

DESCRIBE Customers;

```
MySQL localhost:3306 ssl stud1_test SQL > DESCRIBE Customers;
```

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PRI	NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
City	varchar(25)	YES		NULL	

INSERT INTO Customers (CustomerID,LastName,FirstName,Address,City)

VALUES (96365,'Shah','Hiren','A67,Building10,abc lane','Mumbai');

SELECT * FROM Customers;

```
MySQL localhost:3306 ssl stud1_test SQL > SELECT*FROM Customers;
```

CustomerID	LastName	FirstName	Address	City
96365	Shah	Hiren	A67,Building10,abc lane	Mumbai

INSERT INTO Customers VALUES

(96300,'Singh','Prakesh','B7,House10,abc lane','Mumbai'),

(89790,'Joshi','Ramesh','413,Sadashiv Nagar','Pune');

----Now add approx. 10 customers-----

SELECT * FROM Customers;

```
MySQL localhost:3306 ssl stud1_test SQL > SELECT*FROM Customers;
```

CustomerID	LastName	FirstName	Address	City
10001	Patel	Amit	12, MG Road	Ahmedabad
10002	Kumar	Suresh	45, Nehru Nagar	Delhi
10003	Desai	Meera	78, Park Street	Kolkata
10004	Rao	Anita	23, Jubilee Hills	Hyderabad
10005	Sharma	Vikas	56, Civil Lines	Jaipur
10006	Nair	Arjun	89, Marine Drive	Kochi
10007	Banerjee	Rohit	34, Salt Lake	Kolkata
10008	Mehta	Priya	67, Bandra West	Mumbai
10009	Gupta	Neha	90, Connaught Place	Delhi
10010	Joshi	Kiran	11, FC Road	Pune
89790	Joshi	Ramesh	413, Sadashiv Nagar	Pune
96300	Singh	Prakesh	B7, House10, abclane	Mumbai
96365	Shah	Hiren	A67, Building10, abc lane	Mumbai

ALTER TABLE Customers ADD Email varchar(50);

DESCRIBE Customers;

```
MySQL localhost:3306 ssl stud1_test SQL > ALTER TABLE Customers ADD Email varchar(50);
Query OK, 0 rows affected (0.0211 sec)

Records: 0 Duplicates: 0 Warnings: 0
MySQL localhost:3306 ssl stud1_test SQL > DESCRIBE Customers;
```

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PRI	NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
City	varchar(25)	YES		NULL	
Email	varchar(50)	YES		NULL	

ALTER TABLE Customers ADD Remark varchar(255);

DESCRIBE Customers;

```
MySQL localhost:3306 ssl stud1_test SQL > ALTER TABLE Customers ADD Remark varchar(255);
Query OK, 0 rows affected (0.0154 sec)

Records: 0 Duplicates: 0 Warnings: 0
MySQL localhost:3306 ssl stud1_test SQL > DESCRIBE Customers;
```

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PRI	NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
City	varchar(25)	YES		NULL	
Email	varchar(50)	YES		NULL	
Remark	varchar(255)	YES		NULL	

SELECT * FROM Customers;

```
MySQL localhost:3306 ssl stud1_test SQL > SELECT * FROM Customers;
```

CustomerID	LastName	FirstName	Address	City	Email	Remark
10001	Patel	Amit	12, MG Road	Ahmedabad	NULL	NULL
10002	Kumar	Suresh	45, Nehru Nagar	Delhi	NULL	NULL
10003	Desai	Meera	78, Park Street	Kolkata	NULL	NULL
10004	Rao	Anita	23, Jubilee Hills	Hyderabad	NULL	NULL
10005	Sharma	Vikas	56, Civil Lines	Jaipur	NULL	NULL
10006	Nair	Arjun	89, Marine Drive	Kochi	NULL	NULL
10007	Banerjee	Rohit	34, Salt Lake	Kolkata	NULL	NULL
10008	Mehta	Priya	67, Bandra West	Mumbai	NULL	NULL
10009	Gupta	Neha	90, Connaught Place	Delhi	NULL	NULL
10010	Joshi	Kiran	11, FC Road	Pune	NULL	NULL
89790	Joshi	Ramesh	413, Sadashiv Nagar	Pune	NULL	NULL
96300	Singh	Prakesh	B7, House 10, abc lane	Mumbai	NULL	NULL
96365	Shah	Hiren	A67, Building 10, abc lane	Mumbai	NULL	NULL

UPDATE Customers SET Email='ramesh.joshi@abc.com' , Remark='Regular Buyer' WHERE
CustomerID=89790;

SELECT CustomerID,Address FROM Customers;

MySQL	localhost:3306	ssl	stud1_test	SQL	> SELECT CustomerID ,Address FROM Customers
CustomerID	Address				
10001	12, MG Road				
10002	45, Nehru Nagar				
10003	78, Park Street				
10004	23, Jubilee Hills				
10005	56, Civil Lines				
10006	89, Marine Drive				
10007	34, Salt Lake				
10008	67, Bandra West				
10009	90, Connaught Place				
10010	11, FC Road				
89790	413, Sadashiv Nagar				
96300	B7, House10, abclane				
96365	A67, Building10, abc lane				

1. What is a database? Why do we need a Database Management System such as MySQL?

A database is an organized collection of data that can be easily accessed, managed, and updated.

A Database Management System (DBMS) is software designed to create, maintain, retrieve, and manipulate data stored in a database. MySQL is a widely used relational DBMS.

We need a DBMS because it enables us to:

- Store vast amounts of structured data efficiently.
- Perform fast searches and retrieve records quickly.
- Insert, update, and delete information with ease.
- Minimize data redundancy and duplication.
- Ensure strong data security and access control.
- Maintain data integrity, consistency, and reliability.

2. What is the difference between CREATE DATABASE, SHOW DATABASES, and DROP DATABASE?

Command	Purpose
CREATE DATABASE	Creates a brand-new database in the system.
SHOW DATABASES	Lists all existing databases present in the MySQL server.

DROP DATABASE Permanently removes an existing database along with all its tables and data.

Examples:

CREATE DATABASE College;

SHOW DATABASES;

DROP DATABASE College;

3. What is a table in a database? Explain the purpose of rows and columns in a table.

A table is a structured, two-dimensional format used to store related information within a database using rows and columns.

- Columns represent the specific attributes or properties (fields) of the entity.
- Rows represent individual, complete data records (tuples).

For example:

CustomerID	Name	Age
101	Aman	24
102	Sneha	28

In this table, CustomerID, Name, and Age form the columns, while each entry across forms an individual customer's record (row).

4. What is a primary key? Why was CustomerID selected as the primary key in the Customers table?

A primary key is a column (or set of columns) that uniquely identifies each individual record in a database table.

A primary key must strictly contain unique values and cannot accept NULL values.

CustomerID was selected as the primary key because every customer gets assigned an exclusive identifier. This guarantees accurate record identification, even if multiple customers share the exact same full name.

For example:

CustomerID	Name
101	Aman
102	Aman
103	Sneha

Even though two customers share the name "Aman", their unique CustomerID values prevent confusion.

5. What is the purpose of the VARCHAR and INT data types? Give one suitable example of each.

VARCHAR

VARCHAR is used to store variable-length character strings or textual data.

Example:

Name VARCHAR(50)

Stores text values such as "Aman Verma" or "Sneha Patil".

INT

INT is used to store whole numeric values (integers) without decimal points.

Example:

CustomerID INT

Stores numeric values such as 101, 102, or 500.

6. What is the difference between INSERT, SELECT, UPDATE, and ALTER TABLE?

Command	Purpose
INSERT	Adds new rows/records into an existing table.
SELECT	Fetches and displays data stored in one or more tables.
UPDATE	Modifies existing data values in a table.

ALTER TABLE Modifies the structural definition of a table (e.g., adding, dropping, or modifying columns).

Examples:

INSERT:

INSERT INTO Customers **VALUES** (101, 'Aman');

Adds a new customer entry to the table.

SELECT:

SELECT * FROM Customers;

Retrieves and displays all records from the Customers table.

UPDATE:

UPDATE Customers

SET Name = 'Amit'

WHERE CustomerID = 101;

Changes the name of the customer with ID 101.

ALTER TABLE:

ALTER TABLE Customers

ADD PhoneNumber **VARCHAR**(15);

Adds a new PhoneNumber column to the Customers table.

7. Why is the WHERE clause important in an UPDATE statement? What could happen if we write an UPDATE statement without a WHERE condition?

The WHERE clause is crucial because it filters and specifies the exact rows that need modification.

For example:

```
UPDATE Customers
```

```
SET Name = 'Amit'
```

```
WHERE CustomerID = 101;
```

Only the row matching CustomerID 101 gets updated.

If the WHERE clause is omitted:

```
UPDATE Customers
```

```
SET Name = 'Amit';
```

Every single record in the Customers table will have its Name changed to 'Amit', resulting in accidental data loss and mass overwrite. A proper WHERE condition is essential to target only the intended entries.